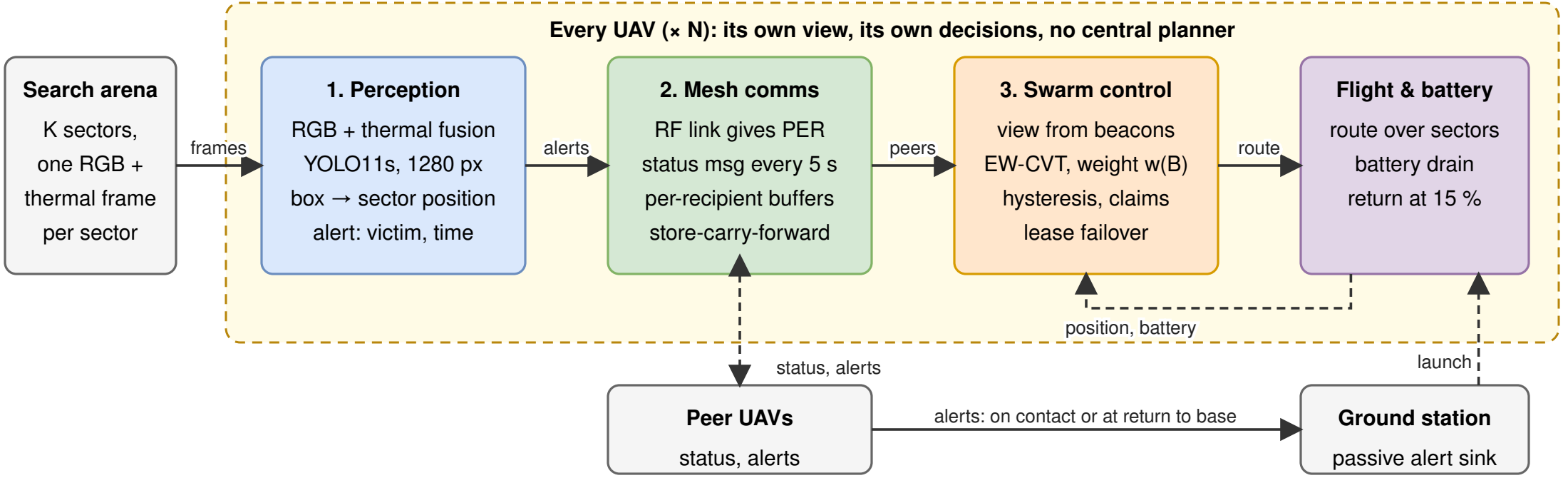
Every UAV (× N): its own view, its own decisions, no central planner
Search arena
K sectors,
one RGB +
thermal frame
per sector
frames
1. Perception
RGB + thermal fusion
YOLO11s, 1280 px
box → sector position
alert: victim, time
alerts
2. Mesh comms
RF link gives PER
status msg every 5 s
per-recipient buffers
store-carry-forward
peers
3. Swarm control
view from beacons
EW-CVT, weight w(B)
hysteresis, claims
lease failover
route
Flight & battery
route over sectors
battery drain
return at 15 %
position, battery
status, alerts
launch
Peer UAVs
status, alerts
alerts: on contact or at return to base
Ground station
passive alert sink